

Nhat (Nick) Trinh

nhattvm1112@gmail.com | New York, NY | [linkedin.com/in/nicktrinh](https://www.linkedin.com/in/nicktrinh) | github.com/nicktrinh

EDUCATION

Fordham University

August 2023 - Expected May 2027

Bachelor of Science in Computer Science, Minor in Math

GPA: 3.6/4.0

- **Coursework:** Data Structures, Computer Algorithms, Artificial Intelligence, Object-Oriented Programming, Operating Systems, Database Systems, Computer Networks, Data Mining, Discrete Mathematics
- **Honors & Awards:** Gold Medal in Informatics, Vietnam Olympic 30/4; 1st Place, Da Nang City Informatics; Faber Award (full-tuition); Dean's List; Sigma Xi Associate Member

EXPERIENCE

Google

May 2026 – August 2026

Software Engineer Intern, Google Compute Engine (GCP)

Seattle, WA

- Architected and launched a release management dashboard to GCE's release platform to digitize the manual and fragmented process, cutting **3-team manual toil** and giving live observability across **12+ releases** per year
- Prevented unsafe rollouts and deployment delays by enforcing a gated path to production with real-time verification of code-review, issue-tracking, and rollout-event state (**Go gRPC, Cloud Spanner**) plus per-phase CLI generation
- Placed **top-13** out of 75 teams at Google's AI-infrastructure hackathon: gave the org's fleet-wide AI agent real-time host rollout understanding through an ADK sub-agent and RPC tool layer to cut incident triage time by **90%**
- Drove team AI adoption by shipping a RAG on-call assistant grounded in curated docs and live fleet state queried through AI agents, and onboarding the team's codebase onto internal semantic code intelligence tool for AI agents

StudyFetch

January 2025 – January 2026

Full-stack Software Engineer Intern

New York, NY

- Co-built a full-stack AI video/podcast SaaS from scratch with Next.js, TypeScript, MongoDB, Prisma, Stripe, Pinecone, and Vercel AI SDK, scaling to **210k+ users** across **129 API routes** and **16 data models**
- Implemented AI streaming chat with Anthropic Claude and vector-based content recommendations via OpenAI embeddings, reducing compute costs by **\$21k/month** through caching, API call batching, and model switching
- Built internal creator portal with magic link auth (JWT, Mongoose), role-based admin dashboard, Google Sheets/Drive API integration, and GitHub Actions CI/CD on Vercel, serving a creator network with **2B+ views**

Fordham Robotics & Computer Vision Lab

February 2024 – Present

Research Assistant

New York, NY

- Published as lead author at IEEE RCC 2025; journal extension under review at IEEE RA-L (arXiv: 2512.09071)
- Developed adaptive thresholding via negative Gaussian mixture statistics for Visual Place Recognition, improving Recall@1 by up to **41.67pp** across 6 descriptors and 16 dataset configurations
- Extended to open-set VPR with online place discovery and reference-only per-place adaptive thresholds; achieved **92–100pp** same environment distractor rejection across datasets vs 77–97pp for Vysotska et al. (ICRA 2025)

Rainscales

June 2024 – August 2024

AI Engineer Intern

Remote

- Engineered YOLO-based computer vision system for workplace monitoring, tracking **1000+** workers' daily alcohol tests with **92% accuracy** using ByteTrack multi-object tracking, line-crossing detection, and bbox overlap analysis
- Optimized model inference by **50%** by evaluating TensorRT and ONNX export pipelines, deploying via NVIDIA Triton Inference Server with CUDA 12.1 GPU acceleration, containerized with Docker for production

PROJECTS

Fordham Registration Agent | *TypeScript, React, Chrome MV3, Anthropic Claude API, IndexedDB* 2026

- **3rd Place, Fordham AI Solutions Challenge 2026** — Chrome MV3 extension embedding a Claude-powered advisor inside DegreeWorks, grounded in live audit data and Banner catalog via reverse-engineered JSON APIs
- Architected PII-safe Anthropic pipeline with placeholder-token redaction, two-tier memory curator (Sonnet 4.6 hard-fact + Haiku 4.5 soft-signal), and prompt caching for **~90% input-token savings** from turn 2

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, TypeScript, Go/Golang, SQL

Frameworks & Libraries: Angular, React, Next.js, Node.js, Express.js, Prisma, Mongoose, PyTorch, TensorFlow, LangChain, Tailwind CSS, Vite, Zustand, Pandas, Scikit-learn, NumPy, OpenCV, ONNX, Stripe

Infrastructure & Tools: Docker, AWS, Google Cloud Platform, Linux, Git, PostgreSQL, MongoDB, Redis, NVIDIA Triton, TensorRT, Vercel, Pinecone, gRPC, Protocol Buffers, IndexedDB, Cloud Spanner

ML & AI: RAG, Vector Databases, AI Agents, Tool Use, Model Context Protocol (MCP), Prompt Caching, Computer Vision, NLP, LLM Operations, Embeddings, OpenAI Whisper, YOLO, ByteTrack, llama.cpp